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Radiation Studies on the ATLAS Link and Control Daughterboard for the HL-LHC Upgrade

1 st Eduardo Valdes Santurio Fysikum, Stockholm University Stockholm, Sweden eduardo.valdes@fysik.su.se ORCID: 0000-0001-9931-2896

2 nd Samuel Silverstein Fysikum, Stockholm University Stockholm, Sweden silver@fysik.su.se ORCID: 0000-0001-7734-7617

3 rd Christophe Clement Fysikum, Stockholm University Stockholm, Sweden christophe.clement@fysik.su.se ORCID: 0000-0003-3122-3605

4 th Christian Bohm Fysikum, Stockholm University Stockholm, Sweden bohm@fysik.su.se

5 th Holger Motzkau Fysikum, Stockholm University Stockholm, Sweden holger.motzkau@fysik.su.se ORCID: 0000-0002-0652-6236

multiplier tubes (PMTs) to the off-detector systems. The on- and off detector electronics are interconnected via a radiation tolerant readout link and control Daughterboard (DB). Each DB is interfaced to a Upgrade TileCal Module that hosts up to 12 PMT channels in the front-end by means of a 400-pin FMC connector. A total of 896 DB will transmit approximately 35 Tbps of data via 5376 optical links to TileCal off-detector Pre- Processor modules. The on-detector systems, which include the DB, has been carefully engineered with redundancy layers and radiation mitigation strategies to maintain performance and reliability with negligible impact on data taking during LHC runs.

Abstract —We present a summary of the radiation studies conducted on the latest version of the ATLAS Tile Calorimeter (TileCal) readout and control Daughterboard (DB), developed for the High Luminosity upgrade of the Large Hadron Collider. The DB functions as a interface hub between the on-detector and off- detector electronics by to handling configuration commands and synchronous timing signals, while simultaneously transmitting monitoring, calibration, and physics data from the front-end electronics. The design incorporates Commercial-off-the-shelf (COTS) devices with field-programmable gate arrays (FPGAs) and CERN custom radiation-hard GBTx ASICs. The COTS were thoughtfully studied and selected for their performance in radiation environments comparable to those expected over the operational lifetime of the collider. Each DB will provide continous readout via two pairs of redundant 9.6 Gbps uplinks continuously receive LHC synchronous timing through a pair of redundant 4.6 Gbps downlinks. The upgraded calorimeter system will include 896 DBs serving a total of 5,376 high-speed optical links. A double-redundant architecture has been implemented to minimize single points of failure, while data integrity is preserved through techniques such as electronic redundancy layers, Triple Mode Redundancy (TMR), Soft Error Management (SEM) and the GBT protocol with Forward Error Correction (GBT-FEC). Extensive testing was performed to assess the impact of total ionizing dose (TID), non-ionizing energy losses (NIEL), and single-event effects (SEE) on system performance under realistic operating conditions. Index Terms —Radiation Tolerant, TID, NIEL, SEU, SEL, SEE

II. THE DAUGHTERBOARD DESIGN

The DB is a read-out link and control board, designed for the upgrade of TileCal [2]. A set of commercial off-the-shelf components and the CERN GBTx custom rad-hard ASIC [4] was selected for the design to achieve good performance for a decade of full functionality in the relatively low radiation environment simulated at the geographical position of the DB in the ATLAS detector. The design follows-up a redundancy layer of two functionally-equivalent halves used on-detector in TileCal modules [1]. Each DB powers four SFP+ modules that drive two 4.8 Gbps downlinks and four 9.6 Gbps uplinks. A pair of CERN GBTx hard ASICs receive LHC synchronized clock signals and configuration commands through the downlinks. Meanwhile, a pair of Kintex Ultrascale (KU) FPGAs distribute clocks and configuration received from the GBTx to the front-end and transmits front-end data and monitoring signals through the uplinks. The ASICs, flash memories and FPGAs of the whole board can be remotely configured through the downlinks. The xADCs of the KU FPGAs monitor temperatures, voltage stability and current from both the DB and the front-end to assure that the on-detector electronics, inaccessible during runs, is well protected against any over-current or Single Event

I. INTRODUCTION

The ATLAS Tile Calorimeter (TileCal) is in charge of identifying hadronic jets and measuring transverse missing energy [1]. TileCal will be impacted by a harsher radiation environment and increased rates of pileup as a consequence of the High Luminosity Large Hadron Collider (HL-LHC) programme. An upgrade of the legacy calorimeter electronics is needed to guarantee full functionality over the 10 year period of the HL-LHC. The upgraded electronics will continuously transmit digitized data from approximately 10000 photo-

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some of the samples being able to reach 21×10^{12} n/cm² and more than 200 Gy without any failures. For SEEs three different tests were performed for studying the SELs and Single Event Upsets (SEUs). A KU+ FPGA was irradiated to 2.2×10^{11} p/cm² with 54 MeV protons, where a SEU rate of 24 SEU/(10^9 p/cm²) and an SEL rate of 2 SEL/(10^9 p/cm²). The presence of SEL in the KU+ FPGA disqualified it for usage on any on-detector design. In the case of the KU FPGA two tests were done with 54 MeV protons and 226 MeV protons where no SEL was observed up to 2.62×10^{11} p/cm² and 3.2×10^{11} p/cm² respectively. As for the SEU rates, we measured 116 SEU/(10^9 p/cm²) and 80 SEU/(10^9 p/cm²). Worst case scenario, the SEU rates represent roughly 5498 SEUs per year out of which approximately 5487 are correctable by the SEM and won't impact the nominal runs due to the use of TMR. The remaining 11 uncorrectable errors per year will be managed by the TMR and can be removed by reloading the FPGA configuration from the FLASH memory during data acquisition stops.

Latch-ups (SEL). Additionally, headers were added for extra flexibility in digital and analog monitoring capabilities. All the components utilized in the design were selected and tested to the simulated radiation levels for the HL-LHC lifetime plus safety factors. The single point of failure mitigation strategies to reduce the impact radiation effects include: redundancy layers, GBTx ASICs, Triple Mode Redundancy (TMR), the Xilinx Soft Error Management (SEM), and GBT-FEC protocol.

III. THE SIMULATED RADIATION ENVIRONMENT

The ATLAS Radiation Tolerant Electronics Proposed Guideline on COTS Lot to Lot Variation [4] was put together by the collaboration as a policy for better consistency in the radiation qualification tasks within the different sub-detector electronics designs [4]. Due to the cylindrical symmetry of the ATLAS detector the radiation requirements were extracted from the radiation levels of the DBs positioned in the most heavily irradiated environments [5]. The minimum Total Ionizing Dose required for qualification of each of the design components is 108 Gy of, with inclusion of safety factors of 1.5 for simulation and 3.0 for lot variation in the components. The Non-Ionizing Energy Losses (NIEL) fluence required for qualification is 13.16×10^{12} n/cm², taking into account safety factors of 1.5 for simulation, 1.3 when using a non-monoenergetic neutron source, and 3.0 to account for lot variation of the components. The total expected fluence for Single Event Effects (SEE) is 4.54×10^{11} h/cm², where a safety factor of 3.0 for simulation was included in the calculations.

V. CONCLUSIONS

The radiation tests performed have qualified the key components of the DB design for TID and NIEL resistance up to the required radiation levels. Additionally, SEE tests have allowed to eliminate the possibility of SEL on the KU FPGA used in the current revision of the design and estimate the impact of SEUs on the data taking for nominal runs. The team plans conducting radiation tests in mixed fields and additional SEU tests with operational firmware to further evaluate the impact of SEU rates on board performance. Additionally, further radiation tests will be required during production phase to account for variance in the batches during the production of components. The tests will follow the ATLAS Radiation Tolerant Electronics Proposed Guideline. The project will produce 930 DBs of which 896 will be installed in ATLAS for the HL-LHC.

IV. DAUGHTERBOARD RADIATION QUALIFICATION STUDIES

A diverse set of radiation campaigns took place were commercial development boards, dedicated boards, and full four assembled DB prototypes that were irradiated to more than 108 Gy and 14×10^{12} n/cm². These tests qualified the Kintex Ultrascale FPGAs, the ProASIC FPGAs, the IS25LP128F, IS25WP256 and IS25LP512 flash memories, DAN217T146CT Schottky diodes, 540BAA100M000BBG oscillators, and INA333AIDRG instrumentation amplifiers. The ProASIC FPGAs and IS25LP128F flash memories showed some after-effects on the re-programmability of the devices that were fully recovered in all devices after annealing and have no impact in the nominal run of the system. Similarly, the impact of ionizing radiation on the threshold voltages the

FDFMA3N109 MOSFETs is acceptable for the digital use of these devices on the design a. In the case of the QUAD oscillators LTC6902IMS it was determined that the inconsistent post irradiation behavior over the sample set was sufficient for disqualifying the device. Four different variants of SFP+ devices from four different companies were tested: AVAGO, CORETEK, FS-Europe and Direktronik. The AVAGO AFBR-709SMZ the CORETEK CT-A000NPP-SB1L-D devices failed to pass the irradiation tests, while both the Direktronik 33-4248 and the FS-Europe SFP-10GSR-85 were qualified with

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